Constructing Language Judgments: Rubric Distillation and Evidence-Grounded Reasoning

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Abstract

Gnomon constructs language judgments through rubric distillation, evidence-grounded reasoning and constrained aggregation. A versioned rubric compiler supplies shared item definitions to an execution harness that elicits observations, verifies source spans and accounts for relation-level credit before constructing criterion scores. Structured extraction targets and candidate-bound critique-and-repair examples connect this inference architecture to verifier-filtered supervision. Completion-focused adaptation preserves example weights across microbatches while chunked vocabulary projection controls logit materialization. IELTS writing assessment provides the implemented application. An archived-input replay demonstrates evidence-to-score construction. Training follows a data-centric, less-is-more regime: 101,175 corpus records across 47 source labels are screened through a 44-check registry, and a leakage-checked core set of 3,447 rows (3,092 train, 355 validation; zero overlap with the 629 held-out reference essays) trains the 27B model (*exact3447-v1*, Qwen3.8-27B) over five epochs to checkpoint-970. Supervised fine-tuning is completed across two runs, 12,417 training rows consumed in total, on 8B multimodal (*adapter_s162*, Qwen3-VL-8B-Instruct, 9,325 rows) and 27B base models; a further 11,466-row curriculum bundle (10,888 train, 578 validation) is registered; and a reward-verified reinforcement path (RLVR with GRPO) is implemented as separate infrastructure. A 2,232-criterion-unit replay frame is prepared for held-out measurement. The contribution is an inspectable integration of judgment construction and learning infrastructure; a current held-out accuracy figure is not yet measured and is not claimed here.

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1 Introduction

The central engineering problem is judgment construction: connecting an evaluative construct to admissible observations, explicit credit rules and a reviewable decision. Gnomon addresses this problem through a rubric-grounded execution harness rather than a single score-prediction prompt. Its intermediate artifacts expose which observations were proposed, which survived verification and how they affected the resulting judgment.

Gnomon separates *observation*, *verification* and *scoring authority*. A model extracts criterion-level evidence; tool and verification components assess it; explicit rules construct criterion scores. Here, *reasoning* means inspectable transformations, not access to a model's hidden cognition.

We use *rubric distillation* to describe the operational translation of evaluative criteria into executable item definitions, structured extraction targets and grounded correction examples. This joins rubric compilation with verifier-filtered supervision; the term names an operational translation procedure, not a knowledge-distillation algorithm in the teacher–student sense. *Reasoning orchestration* names the ordered extraction, verification, credit-accounting and decision stages, including explicitly configured correction branches. IELTS is the implemented testbed for this architecture; transfer to other judgment domains remains to be evaluated.

IELTS Writing uses four criteria per task: Task Achievement (TA) for Task 1 or Task Response (TR) for Task 2, plus Coherence and Cohesion (CC), Lexical Resource (LR), and Grammatical Range and Accuracy (GRA). Five criterion names across tasks do not imply five scores on every essay. General Training Task 1 uses a letter instruction rather than an Academic chart (IELTS, 2023a,b).

Contributions. The implementation connects three systems mechanisms: (1) *rubric-grounded* *judgment construction*, sharing compiled definitions across evidence elicitation, exact-span verification and relation-level credit accounting; (2) *provenance-bound supervision*, binding selected attempts and corrective examples to the candidates they revise; and (3) *resource-aware adaptation*, preserving logical-window example weights and recording supervised-token throughput. Data curation follows the same economy: a less-is-more stance that favours learning signal over corpus volume (Zhou et al., 2023; Li et al., 2023), applied through registered quality checks and leakage exclusion rather than bulk admission. An archived-input replay exercises the scoring path; reconciled population records distinguish corpus inventory, prepared examples and two completed training runs. The contribution is the integration of these mechanisms and their inspectable interfaces.

2 Related work

Interpretable, feature-informed scoring and criterion-specific assessment are established research directions. e-rater V.2 and the broader automated-essay-scoring literature provide the context for this work (Attali and Burstein, 2006; Ke and Ng, 2019). Gnomon studies how data governance, rubric definitions, evidence transport and constrained scoring can be connected through inspectable interfaces.

Automated-scoring evaluation must be tied to intended use, with agreement, error, bias and validity evidence considered together (Williamson et al., 2012). Structured model judging also remains vulnerable to position, verbosity and self-enhancement biases; an inspectable format does not eliminate them (Zheng et al., 2023; Liu et al., 2023).

3 System architecture

Figure 1 shows the connected information flow. The inspected *grade_engine* entry calls *eval_harness*, which assembles evidence for *scorer_3layer*. Rubric content supplies both elicitation and interpretation.

Descriptions distinguish the *default* scoring route, *conditional* branches enabled by configuration, *historical* runs and *prospective* studies. This prevents a component implemented in one configuration from being attributed to every run.

3.1 Rubric compilation and structured elicitation

The rubric specification defines constructs, items, dimensions, graded groups, task variations and evidence requirements. The *compile_rubric* implementation validates it, builds the runtime item bank and assigns a content-based version. Prompt construction and scoring read the same item definitions. Registered dependencies can be updated to the new version; optional additions remain separate from the default bank.

Task metadata selects the extraction route. Its input includes text and, where applicable, an image; the request builder supports task instructions and criterion-specific evidence schemas. The model returns locatable quotations associated with rubric items rather than an essay-band prediction.

The extraction contract restricts model output to observations, while scoring rules and calibration references reside in their own components. This is an information boundary, not a claim that all upstream selection is label-free.

3.2 Evidence verification and judgment construction

The shared scoring consumer combines model-derived item verdicts with deterministic tool signals. It retains distinctions among supported, partially supported, unsupported and unmeasured evidence. Unmeasured evidence and failed presence checks are distinct. In R1, rejection of the legacy flow quotation leaves its presence-rule item UNMET, while the chain and pivot channels return NO_EVIDENCE and are excluded from dimension combination. Zero admissible facts therefore does not automatically mean an unmeasured item.

The default scoring path maps item evidence to criterion dimensions, accounts for graded-ladder structure and applies limiting-dimension aggregation with non-compensatory constraints. A dimension at the top of its measured ladder supplies a lower bound rather than automatically limiting a deeper dimension. Configured psychometric or adopted magnitude-estimation paths can change the base estimate; neither is a universal default. The Rasch branch requires *SCORER_MATH=rasch* and calls *rasch_band* with credited evidence. The complete bank and calibration recipes are not distributed.

For a populated criterion, the order is dimension credit and ladder accounting, limiting aggre-

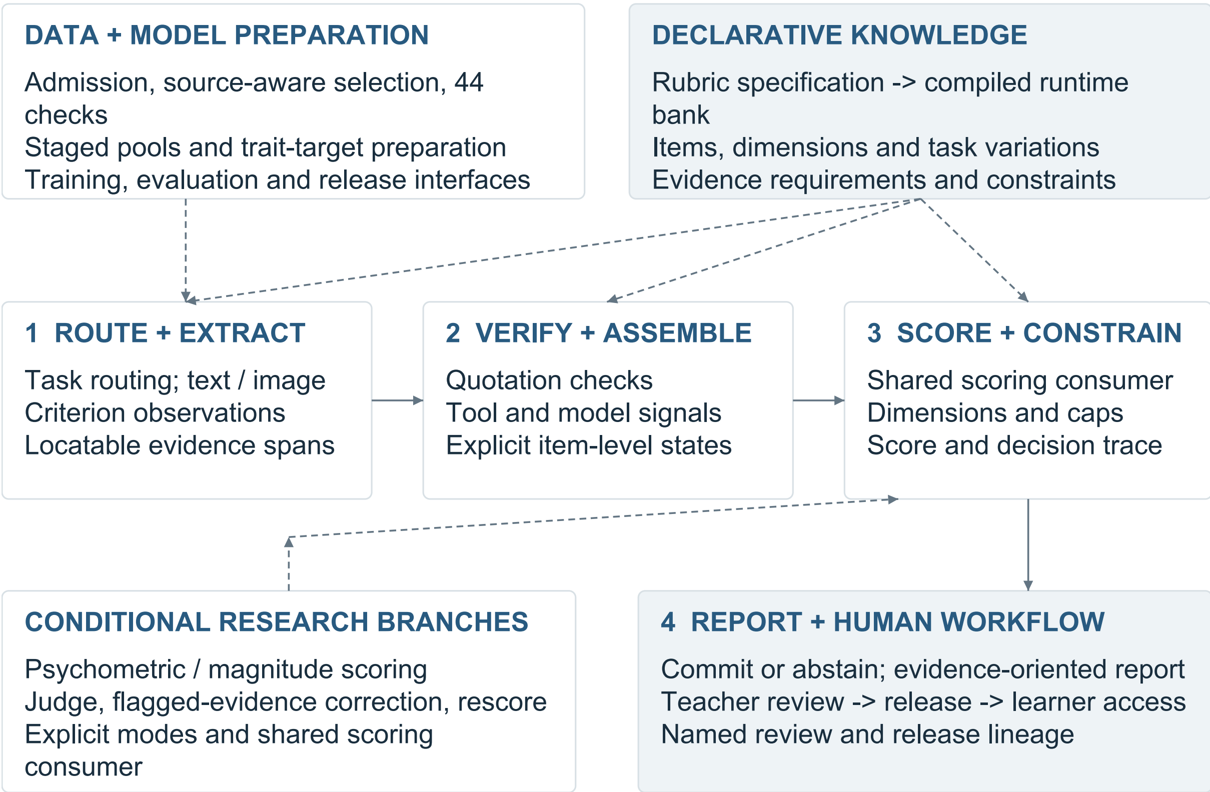

Figure 1: The central assessment path and supporting systems. Dashed links identify configuration or conditional branches; human release remains a separate authority step.

gation, optional base-estimate substitution, corroborated floor/cap application, half-band rounding, then configured low-tier scale and high-tier uncertainty hooks. The evaluation consumer applies deterministic guardrails after scoring; the engine's decision and human-release boundaries remain downstream. The inspected scale and high-tier hooks are disabled/identity by configuration. An empty dimension set takes the earlier fallback in Table 1, rather than traversing the populated combine.

The retained trace lets a reviewer ask separately whether a quotation exists, whether its interpretation is sound, and which scoring condition it affected.

Decision lineage. A review separates *what was observed*, *what earned credit*, and *what was released*. The response and task identify the assessment; the rubric version fixes the item definitions; verified observations feed the scorer; and human review governs learner-facing release. When a reviewer rejects an observation, its affected scoring rule can be located without treating the revised score as new evidence. Shared definitions reduce mismatches between prompting and scoring, while route-level checks establish which links a particular execution preserved.

3.3 Abstention, cross-checks and correction

Disagreement checks can withhold a score when estimates conflict. A single estimate provides no agreement-based confidence evidence, and correlated repeated calls are not independent raters. Evaluation must therefore report coverage alongside error among accepted cases.

The wider engine also contains judge and correction branches. These can cross-check signals, regenerate flagged evidence and rescore it through the shared consumer. Production automatic checks are deterministic-only; model-backed correction and the provenance-bound teaching overlay require explicit modes, callbacks and context. The selected mode determines which branch a report traverses.

The delivery workflow supports teacher review, release, learner access and challenge. Learner-facing feedback must translate a verification reason into a clear diagnosis and manageable action. Its

Table 1: Evidence states are not interchangeable. These are source-bound component semantics, not a claim that every route enforces a universal missing-context refusal.

Evidence state	Implemented consequence
Supported / unsupported	MET and UNMET are measured states. Credit follows the item's role; a measured flaw can activate a configured cap. An unsupported positive claim is not automatically a flaw.
Partially supported	Graduated items can receive intermediate credit. Neutral and prerequisite items do not receive positive partial credit; unmet measured prerequisites can prevent a higher rung from earning credit.
Unmeasured / unavailable	NO_EVIDENCE, omitted tool signals and unmeasured items do not testify as measured failures. A dimension with no measured item is excluded from combination.
No measured dimensions	The scoring component returns its configured floor, then applies its configured scale/uncertainty hooks. In the inspected CC configuration the result is 4.0, not an automatic abstention. This is distinct from having no score estimate.
Missing task context	Task-scoped checks can be skipped when identity is unavailable. A grounded quotation alone cannot certify task fulfilment. Task-aware admission and upstream input controls are separate contracts.
Decision without estimates	The disagreement layer abstains when its estimate list is empty. With one estimate it commits that estimate; this supplies no independent agreement evidence.

helpfulness requires evaluation with teachers and learners.

3.4 Exact-span transport and relation identity

Supported extraction items can use sentence locators and short head/tail anchors instead of reproducing a long quotation. The resolver checks the permitted structure, contiguous sentence range, unique matches and ordering, then slices the original response. It does not repair spelling or rewrite the learner's text. Invalid transport rejects the emission or omits the affected fact with an issue, according to the caller's mode.

The relation verifier works above individual quotations. It checks endpoint grounding, cardinality, permitted item memberships, resolution state, scope and agreement among repeated views. Conflicting identities or inconsistent copies invalidate the relation. An essay-local index retains valid views while assigning a resolved relation's positive credit only to its first eligible item in the immutable rubric order. Repeated presentation of the same relation therefore does not create repeated credit. This is mechanical identity accounting, not a judgement that the relation is semantically correct.

These checks separate quotation recovery, relation identity and interpretation. A reviewer can locate whether an error arose in extraction, verification or scoring; correct quotation recovery alone does not establish a correct interpretation.

3.5 Worked extraction-to-score replay

Trace R1 executes the current shared verification, tool, derived-item, scoring and guardrail consumers on the first existing CC cache record, selected before inspecting its result. The archived response and extraction are bound to file, row and text hashes; the replay records the runtime-bank and owner hashes. It uses the heuristic configuration with no model call, new extraction or provider access.

Table 2 presents selected trace observations, not an exhaustive scoring derivation. The dimension positions and logged base of 5.30 also depend on other verified facts in the archived row; the linker examples alone do not determine 5.5. The complete essay and private bank weights are not supplied for independent numerical reproduction.

The original cached input is unchanged. As a separate negative control, replacing the only retained entity-carry-over quotation with absent text changes that item's verdict from MET to UNMET. This tests the grounding boundary, not whether the original entity interpretation was correct. The archived row has no task identity, so task-scoped completeness is not tested; R1 is a compatibility replay, not a claim of current extraction quality or a full learner workflow. Its 5.5 is one computed criterion score, not an MAE.

4 Data and learning infrastructure

4.1 Quality controls with explicit scope

The *run_full_qa_gate* registry contains 44 distinct checks across seven groups. Figure 3 shows that

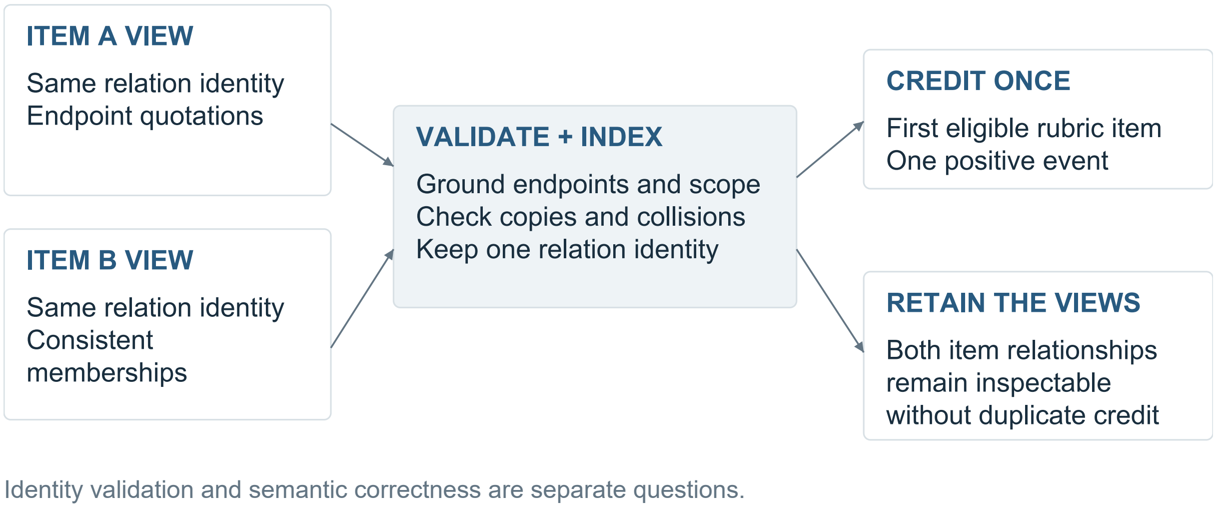

Figure 2: Schematic relation-level accounting. Several item views can refer to one validated relation; scoring credits it once. No learner text or private rubric item is shown.

Table 2: Selected observations from an executed archived-input replay, not an exhaustive scoring derivation or accuracy test. The trace does not establish that the model's semantic interpretations are correct.

Boundary	Observed R1 trace
Source and proposal	The response contains "However" and "On the other hand". The cached model proposes these as linker observations; the discarded "By the way" proposal is not positive evidence.
Grounding and rule	Two retained linker quotations satisfy the configured presence rule: the attempted-cohesion item is MET. A concatenated, noncontiguous quotation proposed for an overlinking flaw contributes zero grounded facts; that flaw rule is UNMET.
Transport rejection	A legacy raw quotation targets a now anchor-only flow item. The resolver records ANCHOR DROP; zero admissible facts leave this presence-rule item UNMET. Relation identifiers are absent in this legacy record; relation-level duplicate credit is not exercised.
Dimension consequence	The resulting coherence position is 5.0; cohesion is 7.0. Cohesion reaches its measured ladder top and becomes a lower-bound observation, while coherence remains limiting.
Score construction	The limiting-weighted base is 5.30. No cap or guardrail changes this case; half-band rounding produces 5.5. Passing this single estimate to the decision component returns COMMIT. This is not a human-approved release.

implemented surface. It describes coverage, not a current pass rate.

Controls address exact and normalized duplication, near-duplicate detection, cross-split overlap, missing or malformed fields, source-label trust, distribution coverage and training-readiness conditions. The implementation distinguishes reporting, review requirements and selected repair actions. The training sub-gate selects eight of the registered checks; it is not equivalent to running the entire quality surface. The eight-check training sub-gate is a fixed, named subset of the full registry chosen for pre-training relevance, not a random or ad hoc sample of the 44.

A central selection API filters by source, task, role, quarantine and evaluation exclusions, then stratifies the selected pools. Each discovered store requires an assigned role and either a named consumer or an explicit deferral. Orchestration links are declared in the registry together with their execution state.

Practice-pool preparation stages all five criterion outputs, validates them with their reader, preserves previous versions and publishes the manifest last. Atomic replacement applies to each file, not to the set as a whole.

4.2 Documented asset and run scale

The September inspection reconciled the inventory, parsed the two prepared SFT files and in-

Figure 3: Static registration counts independently reconciled against the source list. Groups are organizational categories, not statistically independent controls or observed quality improvements.

spected two separate completed-run training manifests. The counts below have different units and overlap; they are not an additive evaluation population.

The current prepared files and the two completed-run checkpoints have separate provenance.

4.3 Admission and supervision construction

Training admission checks identity, usable text, task/criterion compatibility, held-out membership, quarantine and selected quality flags. Academic Task-1 achievement examples require a local image through the shared validity owner; specified score-label patterns are rejected from completions. Named reasons make exclusions countable. No training row reaches the trainer without a hash-bound admission receipt closing first; a receipt that fails to re-resolve blocks model load rather than merely logging a warning. Every write to a locked data partition is paired with an authorizer identity and a before/after row count, so a partial or malformed write is detectable at the row level, not only at the file level. Label-agnostic teaching can skip label-distribution flags while retaining text-quality and integrity checks. Retired per-essay AI-detection scores are not active authorship gates.

A contrastive assembler uses source-score quantiles to select high/low pairs, then emits paired texts with criterion and dimension identifiers but without numeric scores. Selection is label-dependent; only the emitted representation omits absolute scores. A separate weighted absolute-target assembly is maintained as an optional ablation. Supervision format is therefore an inspectable choice rather than an implicit mixture.

Governed library-write tooling records authorisation, snapshots and before/after row counts alongside the mutation. Together with staged pool publication, it provides a lineage record from selected inputs to emitted artifacts.

4.4 Psychometric target preparation

Three measurement concerns are treated separately: targets used during preparation, evidence-to-score calibration, and confidence monitoring.

Trait-target calibration is separate from the contrastive assembler in Section 4.3. Despite its historical MFRM filename, the preparer fits per-trait PCM targets or uses prompt-centred normalization. The latter is a fallback, not a fitted item-response model. Joint rater-severity estimation remains a proposed extension.

Parameter-efficient multimodal adaptation uses established LoRA and QLoRA methods (Hu et al., 2022; Dettmers et al., 2023), with completion-focused supervision. The adapter is a rank-16 LoRA (alpha 16) on the text decoder's linear projections only, with the vision tower and its aligner frozen throughout adaptation. The private curricu-

Table 3: Distinct data-engineering and training records. Source labels are categories, not a count of independent corpora. Preparation rows and historical examples are separate populations.

Record	Recorded scope	Interpretation
Corpus inventory	101,175 records; 47 source labels	Generated inventory across heterogeneous sources.
Prepared SFT data	10,888 train rows; 578 validation rows	11,466 parsed preparation records, prepared and registered.
Earlier 8B run	9,325 examples; 2,134 image-bearing	Qwen3-VL-8B-Instruct; separate non-smoke run; curriculum and weighting enabled.
Second SFT run	3,092 train rows; 355 validation rows	Qwen3.8-27B; leakage-checked core set; five epochs to checkpoint-970; zero overlap with the 629-essay held-out reference set.

Table 4: Implemented calibration-related mechanisms and their roles.

Mechanism	Engineering role	Qualification
Trait-target preparation	Per-trait partial-credit-model estimation where available; prompt-centred normalization fallback.	PCM when fitting is available; normalization otherwise.
Ranking-aware supervision	Continuous trait-target loss combined with adjacent-pair ranking constraints.	Implemented training-loss interface.
Evidence calibration	Conditional Rasch and graded-response modelling, uncertainty estimates and item-fit diagnostics.	Gated research path, not the default scorer.
Scale and source checks	Monotonic score-scale mapping and cross-source item-invariance diagnostics.	Not joint rater-severity estimation.
Confidence monitoring	Prospective decision/outcome joins and confidence-bin summaries.	Separate from grading-accuracy validation.

lum, prompts, remaining adapter hyperparameters and calibration artefacts are not distributed.

4.5 Verifier-filtered supervision and candidate-bound repair

The selection-and-fine-tuning path follows RAFT (Reward rAnked FineTuning) (Dong et al., 2023). The *raft_select* module groups model attempts by essay and ranks eligible attempts by a faithfulness reward. Auxiliary judgement diagnostics are recorded separately and do not select the training target.

For an essay with k passing attempts out of n, the selector keeps a best passing attempt from the mixed region $0 < k < n$. All-pass examples are skipped by this rule; all-fail examples become teacher-rescue candidates. The emitted completion is serialised from the scored extraction, so a caller cannot substitute a different training completion after selection.

Pass predicate and reward. The evaluated object is the structured extraction and its source response. The *verifier_reward* owner requires quote grounding, at least 90% valid fact schemas, no forbidden score-label leakage, sufficient factual coverage for a long response, and passing anti-copy and distinct-evidence checks. When submission-identity arguments are supplied, task/image identity is checked first; legacy callers supplying none skip that contract. A hard-gate failure cannot be rescued by a higher reward.

The ranking reward combines grounding, schema validity, absence of forbidden score labels and evidence coverage, with a small bonus when an attempt includes both retained and discarded observations. Copy detection and distinct-evidence checks determine eligibility only. These tests reject specified extraction failures; semantic accuracy and learner usefulness require separate evaluation.

The optional critique path binds the original candidate, issue set and repair. It is not enabled by the default teaching-loop practice call. A revision must differ from the attempt, satisfy the same gate and improve the base reward:

$$R(e_{\text{revision}}) - R(e_{\text{attempt}}) > 0.$$

STOP responses, invalid bindings, failed revisions and non-positive reward changes are recorded rather than admitted. Each correction is keyed by criterion and mechanism/item identifier, not free-text similarity. Matching keys from first-pass completions serve as anchors; a correction without an

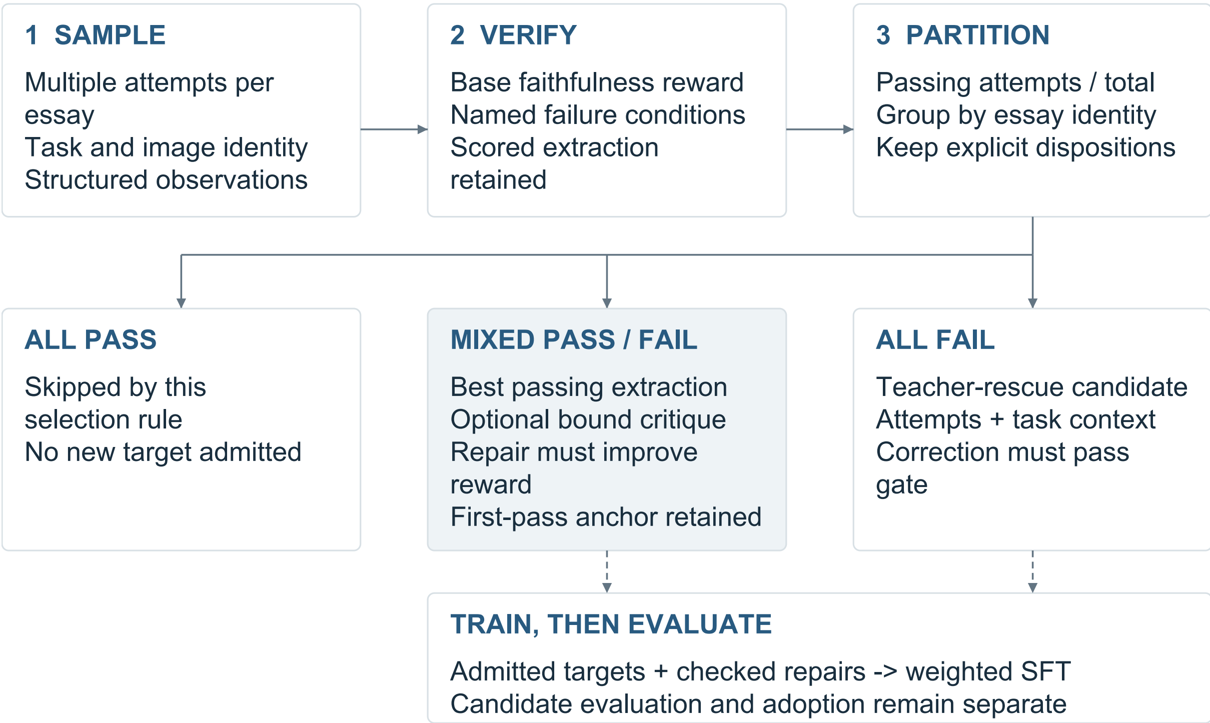

Figure 4: Implemented selection and training interfaces. This is a configured research loop, not a record of a completed production continual-learning campaign. Passing refers to the specified verifier conditions.

anchor is held for review. This preserves first-pass supervision alongside repair examples. Increasing the verifier reward alone does not demonstrate better teaching.

All-fail examples can enter teacher rescue: a teacher supplies a correction, which is checked again against the response and submission identity before acceptance. Training preparation mixes base data, selected attempts and separately supplied verifier-accepted teacher corrections while controlling the weight of self-generated examples. Raw *critic-tuple/v1* records are rejected by the trainer; converted task rows still require promotion and training admission. Thus an accepted critique is not automatically a trainable example.

Candidate adoption. The implemented comparison requires improvement on at least one criterion and no criterion regression beyond its configured tolerance. Missing candidate or baseline assessment receipts stop adoption; a rejected candidate leaves the retained adapter as the next-round seed. The receipt interface requires an isolated, candidate-excluded three-human-median assessment. This is an implemented evaluation requirement, not evidence that such human assessments have been collected.

4.6 Source-conditioned reasoning curriculum

The assembler assigns examples to three phases: method scaffolding, application and generalisation, and consolidation. Source tiers distinguish behaviour-conditioned examples, pristine/examiner examples and legacy material; an essay-identity hash produces deterministic interleaving. Behaviour-conditioned examples concentrate in the early scaffold with some later replay, pristine/examiner examples concentrate in consolidation with earlier interleaving, and legacy examples occupy the application phase. Emitted rows retain source class, tier, phase and weight. This implements a reproducible ordering hypothesis; it does not establish that the ordering improves reasoning or retention.

4.7 Experimental GRPO path

Reinforcement learning with verifiable rewards (RLVR) supplies a checkable reward to policy optimisation; GRPO estimates advantages relative to a group of sampled responses (Shao et al., 2024; DeepSeek-AI et al., 2025). A separate small-model implementation uses TRL's GRPO trainer with Qwen3-0.6B, groups of four responses and a Dr. GRPO loss setting (Liu et al., 2025). Its configuration retains group reward normalisation, uses zero KL coefficient, and applies AdamW updates to LoRA parameters without vLLM.

The smoke's binary reward checks exact relations in synthetic source facts and rejects copying, malformed output and incorrect relations. This is distinct from the extraction-faithfulness reward above. Instrumentation records completion and token identities, group rewards and advantages, checks reward variance and parameter changes, and binds adapter, optimiser, scheduler and random-state checkpoints across a fresh-process resume. Local self-checks cover reward ordering, forged restore identity and checkpoint mutation; they verify restore identity and checkpoint integrity, not learning outcome.

Implementation and evaluation maturity. The present evidence comprises source inspection, the archived-input replay and the recorded run inventory in Table 3. The current training record establishes completed SFT; main-model rollout, loop-training admission and the two-generation GRPO update/resume path are implemented as separate infrastructure. The four-generation small-model path is implemented as separate experimental infrastructure.

5 Training systems and resource accounting

5.1 Completion-focused adaptation

The portable trainer resolves model and precision configuration, checks trainable parameter scope and constructs multimodal batches. Completion-focused supervision masks non-target tokens; per-example weights are carried into the loss interface. Curriculum ordering and length grouping are explicit alternatives: phase order versus grouping supplied sequence lengths to reduce padding. The implementation rejects their simultaneous selection rather than claiming both ordering properties.

5.2 Chunked vocabulary projection

The weighted loss backend projects completion positions through a frozen language-model head in vocabulary chunks. Streaming softmax statistics support full-vocabulary cross-entropy and an entropy term without materialising every sequence position's full logits at once. Backward computation accumulates hidden-state gradients over the chunks; the head must remain frozen.

For a logical optimiser window, the implemented objective is

$$\mathcal{L} = \mathrm{WM}_w(\overline{\mathrm{CE}}) - \beta\,\overline{H}_{\mathrm{completion}},$$

where WM_w is the weighted mean across examples, normalised by the sum of their weights; $\overline{\mathrm{CE}}_i$ averages the completion-token cross-entropy of example i. The entropy mean covers all completion tokens in the window. Common window denominators preserve the intended weighting across microbatches. A materialised reference supplies an arithmetic comparison target. Floating-point reduction order can differ; no new kernel-performance measurement is reported. The objective is implemented as a fused Triton kernel rather than composed from a training library's default cross-entropy path, so no intermediate step materializes uncompressed full-vocabulary logits.

Averaging microbatch means would give a small microbatch the same influence as a large one. Logical-window denominators instead preserve example weights for cross-entropy and token weights for entropy. Vocabulary chunking controls when logits are materialised; it does not change the intended weighting.

5.3 Useful-work denominators

The trainer separately records input tokens, non-padding tokens and completion-loss tokens from windows whose optimiser steps actually applied. These describe batch throughput, useful-token throughput and supervised-target throughput. Keeping them distinct prevents padding from being credited as the same unit of learning work.

Admission, checkpoint evaluation, packaging and serving interfaces connect training to release decisions. Training completion, model selection and serving admission remain separate states. The historical 8B manifest records one execution, not a claim that later trainer components ran in that experiment.

Training runs on rented GPU capacity under a throughput floor gate: a run that falls below its committed floor of 2,500 training tokens per second during an initial smoke phase is halted before the full-scale phase is paid for, rather than absorbed as slow but billed compute. Rental hosts are vetted mechanically before every run, on reliability, verified status, network reachability to the model registry and a standing blocklist of previously failed hosts, rather than chosen on price alone.

5.4 Earlier encoder engineering

Historical ModernBERT work includes per-layer low-rank adapters, fused/separate projection handling and inference-time merging. A DirectML-compatible AdamW path preserves optimiser state and bias correction using supported updates. These are earlier implementations, not dependencies of the current extractor.

6 Evaluation scope and observations

6.1 Study method

This report documents a systematic self-audit of the author's own implementation, its data records and its recorded results. It combines source inspection, reconciliation of population counts, an archived-input scoring replay and reproduction of the aggregate figures.

The initial September 2026 review followed the project registries to 46 source files, then traced admission, relation handling, correction selection and training loss through their callers. A subsequent targeted audit extended the learning account to GRPO, source-conditioned curriculum and their consumer boundaries. A private evidence index links implementation claims to the inspected source versions. Data and run counts were checked against their respective records.

The accompanying *build_figures.py* defines three implementation schematics and one aggregate plot. Its numerical inputs are also exported as *figure_data.json*. Running *python build_figures.py* with ReportLab and Arial/Arial Bold redraws those figures and checks the aggregate arithmetic; the schematic definitions are not measurements. This reproduces the displayed figures, not the original assessment experiment. R1's separate replay script invokes the existing scoring owners on its pinned archived input and emits the resulting trace and checks. It requires the private input and matching runtime, unlike the aggregate figure generator.

7 Discussion

Gnomon makes the score reviewable at several levels: source quotation, rubric interpretation, evidence credit and scoring rule. A teacher or developer can challenge a specific step instead of accepting or rejecting the entire model response. Data selection and correction training use corresponding identifiers, linking assessment review to subsequent learning targets.

The next evaluation should test this design against a direct call to the same model on previously unused, input-complete essays. References and resource budgets should be fixed, with calibration outside the test cohort. Reporting predictions, refusals and exclusions permits comparison of error, coverage and cost. Blinded review of complete feedback, learner comprehension and delayed transfer test teaching outcomes that score agreement alone cannot measure.

8 Conclusion

Gnomon links a compiled rubric, source-grounded observations and explicit scoring rules, with corresponding controls for data preparation and correction training. The report details three reusable mechanisms: exact-span and relation-level credit accounting, candidate-bound repair selection, and completion-weighted training across microbatches. The scoring replay and the recorded run inventory illustrate the implementation and its development record. Matched evaluations will determine where this design improves grading, feedback and delivery cost.

Limitations

The study is retrospective and configuration-specific. It does not include a matched direct-call benchmark, a teacher-preference study or a learner-transfer experiment. Likewise, registered quality checks describe the data pipeline's controls, not the current dataset's pass rate.

The reported implementation represents an intermediate stage of the intended programme of source-conditioned supervision, verifier-filtered practice, teacher rescue, protected candidate evaluation and separately admitted policy optimisation. Present results do not establish an attainable performance ceiling; additional benefit requires controlled evaluation. The historical scoring configuration is separate from this current training programme, and no curriculum-, RAFT- or GRPO-derived capability gain is claimed.

Private datasets, rubric contents, prompts and calibration recipes are not distributed. The aggregate data and figure code reproduce the plots; reproducing an assessment run also requires its inputs, row-level predictions and exact configuration.

Ethical considerations and AI assistance

The intended application is research into formative writing support, not autonomous high-stakes grading or certification. Misinterpreted evidence and unhelpful feedback can mislead learners; source, task and language-background imbalance can produce unequal errors. Human review is a proposed safeguard, not proof those risks have been removed.

The report presents aggregates and short, non-identifying phrases from R1, without distributing the learner response, source record or institutional identifiers. Source licences, consent conditions, examiner qualifications and ethics approvals are outside this inspection and must be established for the intended data sharing, deployment or human study. No official IELTS endorsement is claimed.

Claude and OpenAI Codex assisted drafting, reference checking, figure-generation code and parts of the implementation; the author designed every mechanism described here, and reviewed and owns all committed code. Three figures are programmatic implementation schematics; one plots disclosed aggregates. AI consultation and tool checks are not independent human peer review. The author retains responsibility for final claims, citations and permitted data use.

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